

Notes: Hu (JFE, 2022)
Social interactions and households' flood insurance decisions

Contents

1	Big picture	3
2	Data and measurement	3
2.1	National Flood Insurance Program data	3
2.2	Outcome variables: policies in-force, purchases, and renewals	4
2.3	Flood risk zones and flood-prone policies	4
2.4	Social connectedness data	4
2.5	Shock 1: geographically distant flooding events	5
2.6	Shock 2: FEMA flood-risk-map campaigns	5
3	Empirical specifications	6
3.1	Baseline difference-in-differences for distant floods	6
3.2	Dynamic event-study specification for distant floods	6
3.3	Heterogeneity by connectedness and flood severity	6
3.4	Alternative peer-experience index	6
3.5	Direct flood-risk-map campaign effect	7
3.6	Distant campaign social-interaction specification	7
4	Identification strategy	7
4.1	Core challenge: social effects are hard to identify	7
4.2	Why the NFIP helps identify demand	7
4.3	Why distant shocks help identify social interactions	8
4.4	Parallel trends and pre-trends	8
4.5	Ruling out correlated flood risk	8
4.6	Ruling out migration	8
4.7	Mechanism tests: social learning versus risk aversion	9
4.8	Remaining limitations	9
5	Tables and figures	9
5.1	Table 1: Descriptive statistics of the NFIP	9
5.2	Figure 1: Social connectedness and geographically distant floods	10
5.3	Figure 2 and Table 2: Friends' flood experiences and distant flood insurance purchases	10
5.4	Table 3: Heterogeneous effects of distant floods	11
5.5	Figure 3: Difference between flood occurrences in treated and control counties	11
5.6	Table 4 and Figure 4: Spillovers across insurance products and heterogeneity by flood risk	12
5.7	Table 5: Alternative methodology of estimating the causal effect of social interactions	13
5.8	Table 6 and Figure 5: Intended effects of flood-risk-map campaigns	14

5.9	Table 7 and Figure 6: Distant campaigns and local flood insurance purchases	15
6	Interpretive synthesis	16
7	Short summary	16
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Policy implication	17
8.4	Final takeaway	17

1 Big picture

- **Question.** Do social interactions change households' demand for flood insurance, and can peer exposure to distant flood-risk shocks cause new insurance purchases?
- **Setting.** The paper studies the U.S. National Flood Insurance Program (NFIP), which dominates the market for household flood insurance and offers standardized policies with government-set rates.
- **Main data advantage.** The author uses more than 50 million transaction-level NFIP observations from January 2009 to August 2019, then aggregates to county-month demand measures.
- **Main network measure.** Social links are measured with Facebook's county-pair Social Connectedness Index, transformed into relative friendship probabilities between counties.
- **Identification idea.** The paper studies shocks to geographically distant friends. A flood or flood-risk-map campaign hundreds of miles away should not directly change a county's own flood risk, insurance supply, or insurance prices. Any response in the distant county is therefore interpreted as a social-interaction effect.
- **First quasi-experiment.** Distant friends experience major floods. Counties more socially connected to the flooded area increase flood insurance demand relative to less-connected counties in the same distant state.
- **Second quasi-experiment.** Distant friends experience FEMA flood-risk-map campaigns. Counties more socially connected to the campaign county also increase flood insurance demand.
- **Main result.** Households increase flood insurance purchases by roughly 1–5% after geographically distant friends experience floods or flood-risk campaigns.
- **Persistence.** The effect does not quickly reverse. The additional policies are mostly renewed, suggesting learning rather than a one-period attention spike.
- **Mechanism.** Direct information transmission is possible but likely mild. The strongest interpretation is attention-triggered social learning: friends' flood experiences make flood risk salient and push households to learn about their own exposure.
- **Policy implication.** Social networks can amplify disaster-risk awareness and insurance take-up. This matters because flood insurance is subsidized but underused.

2 Data and measurement

2.1 National Flood Insurance Program data

- The NFIP was created in 1968 to provide household flood insurance where private flood insurance supply is limited.
- NFIP premiums are fixed conditional on risk profiles, especially FEMA-designated flood zones and property characteristics.
- Because premiums do not vary by local market conditions, changes in the number of policies in-force can be interpreted mainly as demand changes rather than supply shifts.
- The paper obtains transaction-level NFIP data from FEMA from January 2009 to August 2019.
- The data include policy effective dates, termination dates, premiums, coverage, deductibles, first policy dates, cancellation dates, community/county identifiers, and property characteristics.
- Policies are renewed annually, so renewals appear as separate transactions.
- The empirical panel begins in January 2010 because the author needs prior transaction history to calculate policies in-force.

Interpretation: The NFIP setting is unusually clean for demand measurement. In many insurance markets, rising sales could reflect either rising demand or a change in prices, brokers, marketing, or underwriting. Here, government-set prices and limited outside options make county-level changes in policy counts closer to shifts in household demand.

2.2 Outcome variables: policies in-force, purchases, and renewals

- **Policies in-force** is a stock variable: the number of effective flood insurance policies in county i and month t .
- **New purchases** is a flow variable: policies that begin in month t .
- **Renewals** are identified using policyholder first-policy dates and annual renewal transactions.
- The main dependent variable normalizes county-level policies in-force to 100 in January 2010:

$$Y_{it} = 100 \times \frac{\text{Policies in-force}_{it}}{\text{Policies in-force}_{i, \text{Jan 2010}}}.$$

- The interpretation of a coefficient is therefore approximately a percentage-point change relative to the county's initial insurance stock.

Interpretation: The normalized stock measure makes small counties and large counties comparable. It also captures persistence: if the effect remains high after the event, the new policies are likely renewed rather than canceled quickly.

2.3 Flood risk zones and flood-prone policies

- The NFIP creates flood hazard maps and classifies areas into flood-risk zones.
- The Special Flood Hazard Area (SFHA) is a zone with at least a 1% annual probability of inundation.
- Flood-prone policies are policies located in SFHAs.
- The share of SFHA policies in a county is used as a proxy for inherent local flood risk in balance and heterogeneity tests.

Interpretation: SFHA shares help separate social-learning effects from genuine local risk differences. If treated and control counties had very different flood-risk profiles, the social-network interpretation would be weaker.

2.4 Social connectedness data

- The paper uses the county-pair Social Connectedness Index from Bailey et al. (2018b), based on anonymized Facebook friendship links.
- The key measure is the relative probability of friendship:

$$p_{ij} = \frac{\text{SCI}_{ij}}{\text{Facebook users}_i \times \text{Facebook users}_j}.$$

- A larger p_{ij} means a Facebook user in county i is more likely to be friends with a Facebook user in county j .

- For a flooded area containing multiple counties $\{j\}_f$, the paper constructs a county-to-event connectedness measure:

$$p_{i,f} = \sum_{j \in \{j\}_f} w_j p_{ij},$$

where w_j is either a population weight or an equal weight.

- The treatment indicator $Connected_i^f$ equals one if county i has above-state-median connectedness to the flooded area.

Interpretation: The state-median threshold is important. Treatment and control counties are compared within the same far-away state, so the design controls for state-level insurance trends, local regulation, and broad state economic conditions.

2.5 Shock 1: geographically distant flooding events

- Flooding events come from the Presidential Disaster Declaration database.
- Flood-related categories include Severe Storm, Hurricane, Flood, Coastal Storm, and Typhoon.
- The sample contains 419 flood-related declarations during the study period.
- The main event-study sample keeps counties at least 750 miles away from the flooded area; robustness uses 500 and 1,000 miles.
- The intuition is that a distant flood affects a household's friends but should not directly change the household's own local flood risk or insurance supply.

Interpretation: The geographical-distance restriction is the central identification device. A flood in one region can plausibly shock the experiences and attention of friends elsewhere without being a local risk shock for those friends' contacts.

2.6 Shock 2: FEMA flood-risk-map campaigns

- FEMA's Risk Mapping, Assessment, and Planning program (Risk MAP) began in 2009.
- The campaign colorized and publicized flood-risk maps, making risk information easier to see and more salient.
- Local governments publicized newly released maps through channels such as open houses and newspaper advertisements.
- The campaign is staggered across counties.
- The direct campaign effect is estimated in targeted counties; the social-interaction effect is estimated in geographically distant counties socially connected to the targeted county.

Interpretation: The campaign shock is useful because it is not a disaster. If distant friends' experiences of information campaigns also increase local insurance demand, the mechanism is less likely to be emotional response to disaster images alone and more likely to involve learning or attention to flood risk.

3 Empirical specifications

3.1 Baseline difference-in-differences for distant floods

For a flooding event f , the baseline specification is:

$$Y_{it}^f = \beta_0 + \beta_1 \text{Connected}_i^f \times \text{PostFlood}_t^f + \beta_2 \text{Connected}_i^f + \beta_3 \text{PostFlood}_t^f + \varepsilon_{it}^f.$$

- Y_{it}^f is the normalized number of policies in-force in county i and month t .
- Connected_i^f equals one if county i is above the state median in connectedness to the flooded counties.
- PostFlood_t^f equals one after the flood event.
- The analysis compares more-connected and less-connected counties in the same distant state.
- Standard errors are clustered at the county level.

Interpretation: The coefficient β_1 is the estimated social-interaction effect of a distant friend’s flood experience on local insurance demand. The identifying assumption is that absent the friends’ flood experience, more- and less-connected distant counties would have followed parallel trends.

3.2 Dynamic event-study specification for distant floods

The dynamic version estimates event-time coefficients:

$$Y_{it}^f = \beta_0 + \sum_k \beta_1^k \text{Connected}_i^f \times 1(t = t^* + k) + \beta_2 \text{Connected}_i^f + \sum_k \beta_3^k 1(t = t^* + k) + \varepsilon_{it}^f,$$

where t^* is the event month and the omitted period is $k = -1$.

Interpretation: This specification checks pre-trends and persistence. A clean design should have near-zero pre-event coefficients; persistent post-event coefficients indicate that additional purchases are renewed.

3.3 Heterogeneity by connectedness and flood severity

The paper runs variants of the baseline regression using:

- top versus bottom quartiles of social connectedness instead of above versus below state median;
- a sample restricted to significant FEMA flood events;
- different distance thresholds and population/equal weighting schemes.

Interpretation: Stronger effects among the most-connected counties and after the most damaging floods support the interpretation that social transmission strength and salience intensity matter.

3.4 Alternative peer-experience index

Following Bailey et al. (2018a), the paper constructs a network-weighted flood-experience index:

$$\text{FriendFlood}_{i,t_1,t_2}^N = \sum_j \theta_{ij}^N \text{Flood}_{j,t_1,t_2},$$

where θ_{ij}^N is the share of county i ’s friends in network N who live in county j , and Flood_{j,t_1,t_2} is the number of floods in county j during the window.

The regression is:

$$\log(\text{Policies})_{it} = \beta \text{FriendFlood}_{i,t-12,t}^{\text{Distant}} + FE_{\text{state} \times \text{time}} + \varepsilon_{it}.$$

Interpretation: This approach uses continuous variation in friends’ flood exposure rather than event-by-event treatment indicators. It is a robustness check that the main result is not produced by one specific event-study construction.

3.5 Direct flood-risk-map campaign effect

For targeted campaign counties, the paper estimates:

$$Y_{it} = \alpha_i + \lambda_t + \beta Campaign_{it} + X_{it}\gamma + \varepsilon_{it}.$$

- α_i are county fixed effects.
- λ_t are year-month fixed effects.
- $Campaign_{it}$ equals one after county i publishes its new flood-risk map.
- Controls include insurance premium, coverage, and flood occurrences.

Interpretation: This is the intended local effect of the FEMA campaign. It shows whether publicizing risk maps changes insurance demand in the target counties before using the campaign as a shock to distant friends.

3.6 Distant campaign social-interaction specification

For a campaign event c , the distant-campaign specification is:

$$Y_{it}^c = \beta_0 + \beta_1 Connected_i^c \times PostCampaign_t^c + \beta_2 Connected_i^c + \beta_3 PostCampaign_t^c + \varepsilon_{it}^c.$$

Interpretation: This mirrors the flood design but replaces disasters with information campaigns. The coefficient measures whether a campaign experienced by distant friends changes local households’ flood-insurance demand.

4 Identification strategy

4.1 Core challenge: social effects are hard to identify

- Peer groups are endogenous: people with similar preferences and risks may become friends.
- Friends may face common shocks: local floods, local campaigns, local economic shocks, and local news can move many households together.
- Insurance purchases reflect both demand and supply in most markets.
- Risk exposure, attention, salience, social learning, and risk aversion may move at the same time.

Interpretation: A simple correlation between a household’s insurance purchase and friends’ purchases would not identify peer effects. The paper therefore avoids local friends and local disasters, and instead uses geographically distant friends’ shocks.

4.2 Why the NFIP helps identify demand

- NFIP rates are centrally set and depend mainly on official risk zones and property characteristics.
- Prices do not adjust to local market conditions or county-level demand shocks.

- Private flood insurance is limited, so households have few substitutes.
- Therefore, changes in policies in-force are interpreted as changes in demand.

Interpretation: The NFIP setting removes a major obstacle in insurance-market research: distinguishing supply shifts from demand shifts.

4.3 Why distant shocks help identify social interactions

- A flood in Boston should not mechanically raise flood risk in California.
- A FEMA map campaign in one county should not directly change flood-zone information in a county hundreds of miles away.
- The same-state comparison among distant counties controls for state-level trends.
- The connectedness split captures differential exposure to friends in the shocked area.

Interpretation: The design is a network exposure design. Treatment is not local risk exposure; treatment is social proximity to people exposed to a distant shock.

4.4 Parallel trends and pre-trends

- Event studies for floods and campaigns show no differential pre-trends between more-connected and less-connected counties.
- Post-event coefficients rise after the shock and remain positive.
- The visual pre-trend evidence is crucial because connected counties may differ in baseline characteristics.

Interpretation: If the treated counties were already trending upward before the shock, the results would be less credible. The figures show the increase begins after event time zero.

4.5 Ruling out correlated flood risk

- The paper checks whether treated counties also experience more local floods around the event.
- Figure 3 shows no differential flood occurrences between treated and control counties.
- The paper also compares inherent flood risk using SFHA shares.

Interpretation: This addresses the most obvious non-causal explanation: perhaps more-connected counties are simply riskier or experience local floods at the same time. The evidence does not support that interpretation.

4.6 Ruling out migration

- Flood victims may move to counties where friends or family live.
- If migrants buy flood insurance after moving, that could mimic a peer effect.
- The paper uses IRS county-to-county migration data.
- Long-distance migration from flooded areas is very small: average 3.6 households per 750-mile-away county within one year, median zero.
- The difference between more- and less-connected counties is statistically insignificant.

Interpretation: Migration cannot plausibly explain a persistent county-level increase in insurance demand.

4.7 Mechanism tests: social learning versus risk aversion

- The proposed mechanism is social learning, especially indirect attention-triggered learning.
- If the channel were generalized risk aversion, distant flood exposure should increase purchases of other insurance products too.
- Table 4 shows no effect on health or earthquake insurance.
- Placebo tests using non-flood disasters also show no similar effect.

Interpretation: The behavior appears product-specific: people respond by buying flood insurance, not by broadly increasing all insurance demand. That points to flood-risk learning or salience rather than a generic risk-aversion shock.

4.8 Remaining limitations

- The paper cannot observe the content of all conversations among friends.
- Facebook connectedness proxies for broader online and offline social ties, but it is not a complete map of social interaction.
- The paper cannot fully separate direct information from attention-triggered learning.
- Welfare is not fully evaluated because higher take-up may involve subsidies and government fiscal costs.

Interpretation: The causal evidence for a social-interaction effect is strong, but the exact micro-channel and welfare implications require additional data.

5 Tables and figures

5.1 Table 1: Descriptive statistics of the NFIP

Read:

- In an average month, the NFIP has 5.29 million policies in-force.
- The program collects about \$3.32 billion in premiums and provides \$1.26 trillion in coverage.
- The average annual premium per policy is \$628; the average coverage per policy is \$238,000.
- About 52.2% of policies are in Special Flood Hazard Areas.
- County-level variation is large: the mean county has 1,766 policies, but the median has only 120.

Interpretation: The NFIP is economically large, but participation is highly uneven across counties. This cross-sectional variation makes county-month analysis meaningful but also requires careful normalization and controls.

Economic message: The table establishes that flood insurance demand is low and heterogeneous. The setting is large enough for policy relevance and granular enough for empirical identification.

Table 1

Descriptive statistics of the NFIP.

This table presents descriptive statistics that characterize the National Flood Insurance Program (NFIP) data in my sample from January 2010 to August 2019. Panel A summarizes the time-series variations for the entire program. *Policies in-force* is the number of effective insurance policies in a given month. *Premium* is the total dollar amount of premiums collected from the policies in-force in a given month. *Coverage* is the total dollar amount of coverages for the policies in-force in a given month. *Flood-prone policies* is calculated as the number of policies in the Special Flood Hazard Area (SFHA) over the total number of policies. The NFIP creates risk maps and designates flood zones, and the SFHA is defined as the area that has a 1-percent or higher probability to be inundated in any given year. Panel B summarizes the cross-sectional variations of the data at the county level.

Panel A: Nationwide time-series variation					
	mean	s.d.	25th pctl.	50th pctl.	75th pctl.
<i>Policies in-force (m)</i>	5.29	0.22	5.08	5.32	5.51
<i>Premium (\$b)</i>	3.32	0.13	3.26	3.30	3.41
<i>Coverage (\$b)</i>	1.26	0.03	1.24	1.26	1.28
<i>Premium per policy (\$)</i>	628	35.8	599	647	654
<i>Coverage per policy (\$k)</i>	238	12.1	229	239	249
<i>Flood-prone policies (%)</i>	52.2	3.8	49.6	52.9	55.7

Panel B: County-level cross-sectional variation					
	mean	s.d.	25th pctl.	50th pctl.	75th pctl.
<i>Policies in-force</i>	1,766	12,563	31	120	437
<i>Premium (\$k)</i>	1,108	6,251	21	88	325
<i>Coverage (\$m)</i>	421	3,023	4.8	20	83
<i>Premium per policy (\$)</i>	754	358	554	697	876
<i>Coverage per policy (\$k)</i>	185	68	137	185	232
<i>Flood-prone policies (%)</i>	53.3	23.8	37.8	55.2	70.8

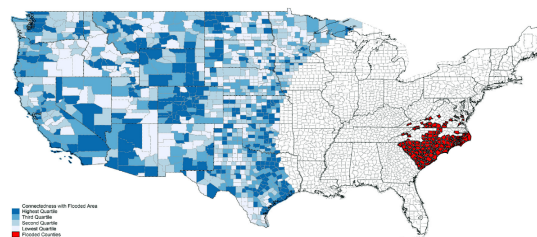


Fig. 1. Social connectedness and geographically distant floods. This figure shows one specific example to illustrate the empirical design of my quasi-experiment. Hurricane Florence hit South Carolina, North Carolina, and Virginia in September 2018. The flooded counties are red-shaded on the map. The

5.2 Figure 1: Social connectedness and geographically distant floods

Read:

- The figure maps Hurricane Florence's flooded counties in the Southeast and the social connectedness of distant counties.
- Only counties at least 750 miles away are considered.
- Blue shading measures stronger or weaker social connectedness to the flooded area.

Interpretation: The map shows the logic of the research design. Counties far from the actual flood differ in social proximity to the flooded region. This generates variation in social exposure holding direct physical exposure far away.

Economic message: The empirical comparison is not between flooded and non-flooded places. It is between distant places with stronger versus weaker social ties to the flooded places.

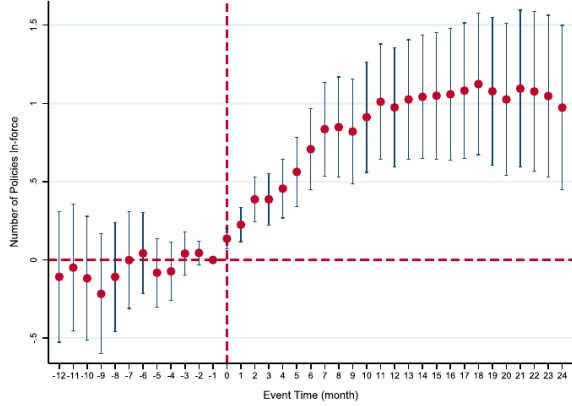
5.3 Figure 2 and Table 2: Friends' flood experiences and distant flood insurance purchases

Read:

- Figure 2 shows no clear pre-trend before the distant flood.
- After the flood, the number of policies in-force rises gradually and remains elevated.
- Table 2 estimates a 0.86–1.54 percentage-point increase depending on weighting and distance thresholds.
- The baseline population-weighted 750-mile estimate is 1.11; this corresponds to about a 0.94% increase over the mean at event time zero.
- Restricted 2014–2017 estimates are similar, addressing concerns that Facebook connectedness is a 2016 snapshot.

Interpretation: The figure and table jointly support the main causal result. The event-study timing shows post-event movement, while the table shows robustness across sample periods and distance definitions.

Economic message: Friends' flood experiences affect local insurance demand even when the flood is geographically distant. Social networks transmit disaster salience or learning beyond affected regions.



experiences on insurance purchases. This figure shows the dynamic effects of geog
(a) Original Figure 2: Friends' flood experiences on insurance purchases.

Table 2

Distant floods and local flood insurance purchases.

This table shows results from the event study design: $Y_{it}^f = \beta_0 + \beta_1 * Connected_i^f \times PostFlood_t^f + \beta_2 * Connected_i^f + \beta_3 * PostFlood_t^f + \epsilon_{it}^f$. For a given flooding event f and the associated flooded counties $\{j\}_f$, county i 's social connectedness to $\{j\}_f$ is measured by the relative probability of Facebook friendship $p_{i,f} = \sum_{j \in \{j\}_f} w_j * p_{i,j}$, where $p_{i,j}$ is the county-by-county probabilities obtained from Bailey et al. (2018b). w_j represents population-weighting or equal-weighting scheme. $Connected_i$ is a binary variable indicating if county i is socially connected with the flooded area, which is defined as having a value of $p_{i,f}$ above the state-median. The analysis sample consists of only counties that are geographically distant to the flooded area. Three different choices of distance threshold are considered: 500, 750, and 1,000 miles. $PostFlood_t$ is a binary variable indicating post-flood periods. The dependent variable Y_{it} measures the number of flood insurance policies in-force (with January 2010 normalized to 100) in county i in month t . Panel A uses the full sample period from January 2010 to August 2019. Panel B uses a restricted sample period from January 2014 to December 2017. Standard errors are clustered at the county level and presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
Panel A: Full Sample Period 2010–2019				
<i>Connected × PostFlood</i>	1.11*** (0.20)	0.90*** (0.20)	0.86*** (0.19)	1.54*** (0.24)
Observations	23,369,392	23,369,392	32,114,276	16,335,272
R-squared	0.066	0.046	0.068	0.063
Panel B: Restricted Sample Period 2014–2017				
<i>Connected × PostFlood</i>	1.28*** (0.28)	0.86*** (0.26)	0.77*** (0.23)	1.72*** (0.35)
Observations	8,760,517	8,760,517	12,028,168	6,019,329
R-squared	0.060	0.035	0.061	0.058
Connectedness Weight	PW	EW	PW	PW
Distance Threshold	750 miles	750 miles	500 miles	1000 miles

(b) Original Table 2: Distant floods and local flood insurance purchases.

5.4 Table 3: Heterogeneous effects of distant floods

Read:

- Panel A compares top versus bottom quartiles of social connectedness.
- Effects are larger than in the median-split baseline: 1.74–2.98 percentage points.
- Panel B restricts to significant floods.
- Significant floods produce much larger estimates: 3.08–5.95 percentage points.

Interpretation: The heterogeneity aligns with the mechanism. If social networks transmit information or attention, stronger network ties should generate larger effects. If salience matters, more damaging floods should also generate larger effects.

Economic message: Social transmission is not a uniform background pattern. It is stronger when social links are stronger and when the distant shock is more severe.

5.5 Figure 3: Difference between flood occurrences in treated and control counties

Read:

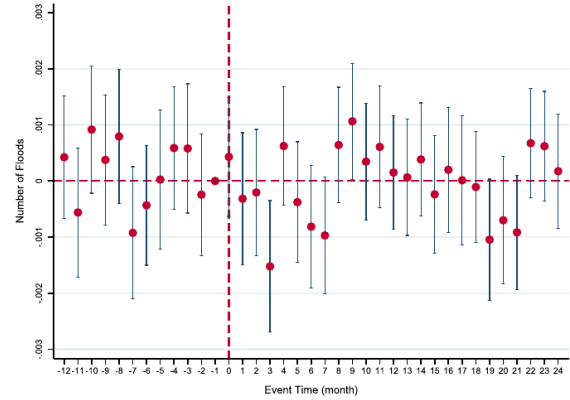
- The figure plots event-time differences in local flood occurrences between treated and control counties.
- Coefficients fluctuate around zero before and after the distant flood.
- Confidence intervals generally include zero.

Table 3

Heterogeneous effects of distant floods.

This table shows results from the event study design: $Y_{it}^f = \beta_0 + \beta_1 * Connected_i^f \times PostFlood_t^f + \beta_2 * Connected_i^f + \beta_3 * PostFlood_t^f + \epsilon_{it}^f$. Panel A and B consider two deviations from the baseline specifications presented in Table 2. All variables are defined as per Table 2 except that in Panel A, $Connected_i$ is defined as a binary variable that equals 1 if county i has a connectedness measure in the top quartile of the state and equals 0 if county i has a connectedness measure in the bottom quartile of the state; Panel B only includes the 18 significant flooding events defined by FEMA. Standard errors are clustered at the county level and presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

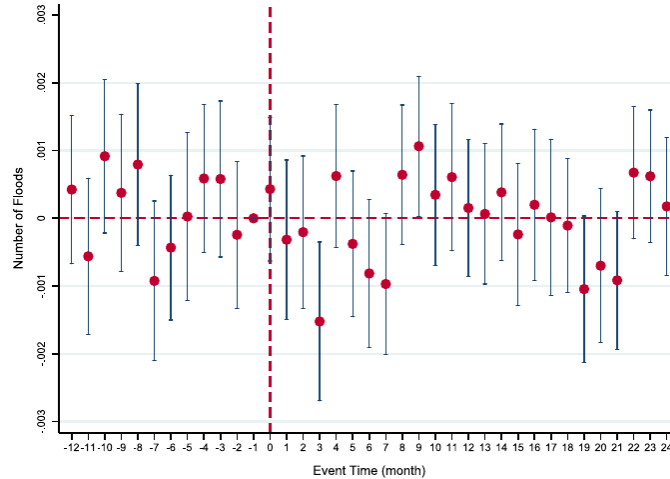
	(1)	(2)	(3)	(4)
Panel A: Top vs. Bottom Quartiles of Connectedness				
<i>Connected</i> × <i>PostFlood</i>	2.20*** (0.30)	1.74*** (0.30)	1.91*** (0.30)	2.98*** (0.37)
Observations	11,965,523	11,965,523	12,576,973	8,101,031
R-squared	0.14	0.09	0.14	0.13
Panel B: Subsample of Significant Floods				
<i>Connected</i> × <i>PostFlood</i>	3.08** (1.43)	4.57*** (1.44)	4.12** (1.64)	5.95*** (2.02)
Observations	688,352	688,352	1,107,478	381,157
R-squared	0.09	0.04	0.10	0.08
Connectedness Weight	PW	EW	PW	PW
Distance Threshold	750 miles	750 miles	500 miles	1000 miles



1 occurrences in the treated and control counties. This figure shows the dynamic effect

Interpretation: This figure directly tests a correlated-shock concern. If more-connected counties also experienced more local floods, the main result could reflect local risk rather than social interaction. The figure shows this is not the case.

Economic message: The increase in insurance demand is not driven by treated counties facing more local flood events.



1 occurrences in the treated and control counties. This figure shows the dynamic effect

Original Figure 3: Difference between flood occurrences in treated and control counties.

5.6 Table 4 and Figure 4: Spillovers across insurance products and heterogeneity by flood risk

Read:

- Table 4 shows no significant spillover to health insurance or earthquake insurance.
- The placebo using non-flood disasters also does not reproduce the main effect.

- Figure 4 shows larger social-interaction effects in higher flood-risk subsamples.
- The effect is small in the lowest-risk counties and stronger in riskier counties.

Interpretation: Table 4 rules out a broad risk-aversion story. If households simply became more risk averse after seeing friends’ disasters, they should buy more insurance generally. Figure 4 suggests attention-triggered learning: the effect is strongest where households can discover meaningful local risk once attention is drawn to flooding.

Economic message: Social interactions appear to target relevant risk, not just general anxiety. The mechanism is closer to flood-risk learning than to broad insurance demand.

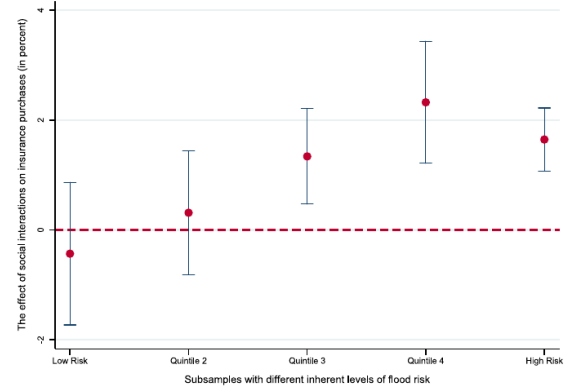
Table 4

Spillover effects across insurance products.

This table shows results from the event study design as described in Table 2. Panel A and B construct the analysis around flooding events and examine changes in health and earthquake insurance purchases: $Y_{it}^f = \beta_0 + \beta_1 * Connected_i^f * PostFlood_t^f + \beta_2 * Connected_i^f + \beta_3 * PostFlood_t^f + \epsilon_{it}^f$. In Panel A, the dependent variable Y_{it} measures the health insurance coverage (in percent) for county i in year t . In Panel B, Y_{it} measures the number of earthquake insurance policies in-force (with January 2010 normalized to 100) in county i in year t (earthquake insurance data is only available for Missouri counties). Panel C considers a placebo test, which constructs the event study around non-flood natural disasters: $Y_{it}^f = \beta_0 + \beta_1 * Connected_i^f * PostDisaster_t^f + \beta_2 * Connected_i^f + \beta_3 * PostDisaster_t^f + \epsilon_{it}^f$. In Panel C, Y_{it} measures the flood insurance demand in county i in month t , same as in Table 2. Standard errors are clustered at the county level and presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
Panel A: Dependent Variable = Health Insurance				
<i>Connected</i> × <i>PostFlood</i>	−0.012 (0.016)	0.009 (0.016)	−0.011 (0.014)	−0.015 (0.020)
Observations	2,527,863	2,527,863	3,440,817	1,671,286
R-squared	0.013	0.010	0.013	0.013
Panel B: Dependent Variable = Earthquake Insurance				
<i>Connected</i> × <i>PostFlood</i>	0.096 (0.460)	0.247 (0.435)	0.134 (0.407)	0.502 (0.523)
Observations	81,729	81,729	126,783	44,778
R-squared	0.005	0.007	0.003	0.010
Panel C: Placebo Using Non-flood Disasters				
<i>Connected</i> × <i>PostDisaster</i>	−0.115 (0.277)	−0.340 (0.277)	−0.055 (0.247)	0.217 (0.307)
Observations	13,850,539	13,850,539	17,493,201	9,835,717
R-squared	0.041	0.036	0.033	0.035
Connectedness Weight	PW	EW	PW	PW
Distance Threshold	750 miles	750 miles	500 miles	1000 miles

(a) Original Table 4: Spillover effects across insurance products.



(b) Original Figure 4: Heterogeneous effects by flood risk. This figure shows the coefficient estimate of $\beta_1 * Connected_i^f + \beta_3 * PostFlood_t^f + \epsilon_{it}^f$, using subsamples. Subsamples comprise states that are defined as in Table 2. Standard errors are clustered at the county level.

5.7 Table 5: Alternative methodology of estimating the causal effect of social interactions

Read:

- The table uses a continuous friend-flood-exposure measure rather than event-by-event treatment indicators.
- Panel A shows that each additional flood experienced by geographically distant friends increases local insurance demand by about 1.1–1.3% in the baseline window.
- Panel B shows larger effects for significant flood experiences, around 3.3–6.7% depending on window and distance.

Interpretation: This table shows that the main effect is not an artifact of the binary connectedness measure. Network-weighted exposure produces a similar conclusion: distant friends’ flood experiences increase

local insurance demand.

Economic message: Social learning operates on a gradient. More flood exposure in the social network creates more insurance demand.

Table 5
Alternative methodology of estimating the causal effect of social interactions.
This table shows results from regression: $\log(\text{Policies})_{i,t} = \beta * \text{FriendFlood}_{i,t-k,t}^{\text{Distant}} + FE_{\text{state} \times \text{time}} + \epsilon_i$. Following the methodology proposed by Bailey et al. (2018a), $\text{FriendFlood}_{i,t-k,t}^{\text{Distant}}$ measures the average flood experience of a county i 's social network N between $t-k$ and t . $\text{FriendFlood}_{i,t-k,t}^{\text{Distant}}$ is calculated as the weighted average as $\sum_j \theta_{ij}^N * \text{Flood}_{j,t-k,t}$, where θ_{ij}^N is share of county i 's friends in network N who live in county j , and $\text{Flood}_{j,t-k,t}$ is the number of floods in county j between $t-k$ and t . A geographically distant network $N = \text{Distant}$ is a set of counties that are certain miles away from county i . Columns 1 through 4 show results of using 750 miles as the threshold; columns 5 through 7 use 250, 500, and 1,000 miles, respectively. The measurement window (i.e., k) of floods takes values of 3, 6, 12, or 24 months. $FE_{\text{state} \times \text{time}}$ are the state $\times$ time fixed effects. Standard errors are clustered at the county level and presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Geographically Distant Friends' Flood Experiences							
$\text{FriendFlood}_{i,t-k,t}^{\text{Distant}}$	0.013*** (0.002)	0.012*** (0.002)	0.011*** (0.002)	0.013*** (0.002)	0.009*** (0.002)	0.011*** (0.003)	0.012*** (0.003)
Observations	321,511	349,443	340,120	284,729	321,511	321,511	321,511
R-squared	0.42	0.42	0.42	0.42	0.42	0.42	0.42
Panel B: Geographically Distant Friends' Significant Flood Experiences							
$\text{FriendFlood}_{i,t-k,t}^{\text{Distant}}$	0.042*** (0.004)	0.041*** (0.005)	0.041*** (0.004)	0.033*** (0.003)	0.043*** (0.005)	0.067*** (0.006)	0.042*** (0.004)
Observations	321,511	349,443	340,120	284,729	321,511	321,511	321,511
R-squared	0.42	0.42	0.42	0.42	0.42	0.42	0.42
Distance Threshold	750 miles	750 miles	750 miles	750 miles	250 miles	500 miles	1000 miles
Flood Window	12 months	3 months	6 months	24 months	12 months	12 months	12 months

Original Table 5: Alternative methodology of estimating the causal effect of social interactions.

5.8 Table 6 and Figure 5: Intended effects of flood-risk-map campaigns

Read:

- Table 6 estimates the direct effect of FEMA map campaigns in targeted counties.
- The campaign increases policies in-force by roughly 20–21 normalized points in most specifications.
- The interaction with high intra-county SCI is positive, suggesting stronger local amplification in more connected counties.
- Figure 5 shows a sharp post-campaign increase and no meaningful pre-trend.
- By the end of the first year, policies in-force increase by about 32.8 normalized points, or roughly 30.6% over the mean at event time zero.

Interpretation: The campaign works directly in targeted counties. That validates the campaign as a meaningful risk-awareness shock before using it as an indirect shock to distant friends.

Economic message: Public information campaigns can sharply raise take-up. Local social connectedness appears to amplify the direct campaign effect.

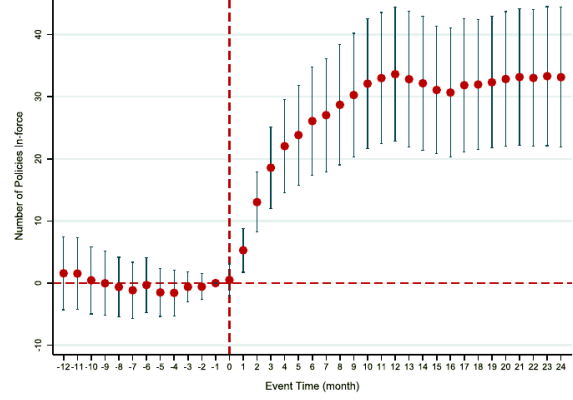
Table 6

The intended effects of flood-risk-map campaigns.

This table shows results from the two-way fixed-effect regression: $Y_{it} = \alpha_i + \lambda_t + \beta * Campaign_{it} + X_{it} + \gamma * Campaign_{it} \times HighSCI_i + \epsilon_{it}$. The panel covers 3,053 counties from January 2010 to August 2019. The dependent variable Y_{it} measures the number of flood insurance policies in-force (with January 2010 normalized to 100) in county i in month t . The main explanatory variable $Campaign_{it}$ is a binary variable indicating if county i has publicized its new flood risk maps at time t ; this term is set to zero for the control counties without campaigns. α_i and λ_t are the county and year-month fixed effects. $HighSCI_i$ is a binary variable that equals one if county i has an above-median value of the intra-county relative probability of friendship. X_{it} are the covariates: $Premium_{it}$ is the average premium per policy (in \$); $Coverage_{it}$ is the average coverage per policy (in \$k); $Flood_{it}$ is the number of flood occurrences. Standard errors are clustered at the county level and presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
<i>Campaign</i>	21.394*** (5.661)	21.266*** (5.668)	20.570*** (5.143)	5.922* (3.183)
<i>Premium</i>		-0.007 (0.007)	-0.005 (0.013)	-0.007 (0.007)
<i>Coverage</i>		-0.058 (0.049)	0.005 (0.103)	-0.059 (0.050)
<i>Flood</i>		0.987* (0.553)	1.472 (0.905)	1.096* (0.566)
<i>Campaign × HighSCI</i>				36.334*** (12.349)
Observations	347,852	347,852	176,251	347,852
R-squared	0.69	0.69	0.75	0.69
Include never-treated	Y	Y	N	Y

(a) Original Table 6: Intended effects of flood-risk-map campaigns.



(b) Original Figure 5: Impact of flood-risk-map campaigns.

5.9 Table 7 and Figure 6: Distant campaigns and local flood insurance purchases

Read:

- Table 7 shows that distant friends' campaign experiences increase local flood insurance demand.
- The baseline 750-mile estimate is 1.21 normalized points, approximately a 1.01% increase relative to the event-time-zero mean.
- The effect is robust to 250-, 500-, and 1,000-mile distance thresholds.
- Figure 6 shows no pre-trend and a gradual, persistent post-campaign increase.
- The increase is slower than the direct campaign response in Figure 5.

Interpretation: The indirect campaign result is important because it is not driven by flood damage. It shows that social interactions can transmit attention to flood risk even when the original shock is information provision rather than disaster experience.

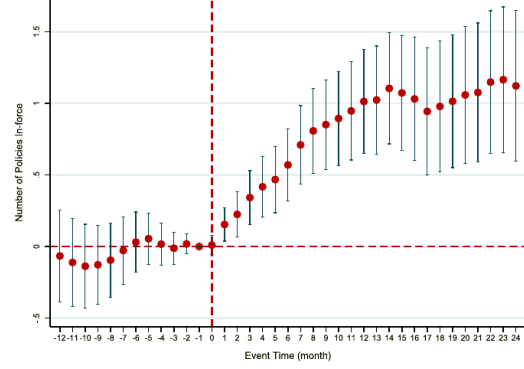
Economic message: Risk communication has network spillovers. A local FEMA campaign can affect households far away through social ties.

TABLE 7

Table 7
Distant campaigns and local flood insurance purchases.
This table shows results from the event study design: $Y_{it} = \beta_0 + \beta_1 \times \text{Connected}_i \times \text{PostCampaign}_t + \beta_2 \times \text{Connected}_i + \beta_3 \times \text{PostCampaign}_t + \epsilon_{it}$. For a given campaign event c and the associated campaign county $\{j\}$, county i 's social connectedness to $\{j\}$, is measured by the relative probability of Facebook friendship p_{ij} obtained from Bailey et al. (2018b). Connected_i is a binary variable indicating if county i is socially connected with the campaign county, which is defined as having a value of p_{ij} above the state-median. The analysis sample consists of only counties that are geographically distant to the campaign county. Three different choices of distance threshold are considered: 500, 750 and 1000 miles. PostCampaign_t is a binary variable indicating post-campaign periods. The dependent variable Y_{it} measures the insurance demand in county i in month t , which is defined as per Table 2. Standard errors are clustered at the county level and presented in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	(1)	(2)	(3)	(4)
Connected $\times$ PostCampaign	1.21*** (0.18)	0.93*** (0.11)	1.30*** (0.16)	1.58*** (0.22)
Observations	56,017,387	107,644,567	83,192,216	34,073,468
R-squared	0.03	0.03	0.03	0.03
Distance Threshold	750 miles	250 miles	500 miles	1000 miles

(a) Original Table 7: Distant campaigns and local flood insurance purchases.



(b) Original Figure 6: Friends' campaign experiences on insurance purchases. This figure shows the dynamic effects of campaign experiences on insurance purchases.

6 Interpretive synthesis

- **Why distant friends matter.** The paper shows that flood risk is socially salient. Even when a flood is geographically irrelevant to a household's own property, it can prompt households to reconsider their own exposure.
- **Why the effect is persistent.** The persistence is consistent with learning. Once households search for their own risk or learn about NFIP products, the updated belief or awareness can have a durable effect.
- **Why campaign evidence matters.** Floods may trigger emotional responses. Campaigns are information shocks. The fact that both shocks transmit through social networks supports a broader social-learning channel.
- **Why product-specific tests matter.** No spillover to health or earthquake insurance weakens the view that social exposure simply raises generic fear or risk aversion.
- **Why policy can use networks.** Information campaigns may have multipliers when targeted to socially connected counties or groups.

7 Short summary

- The paper studies social interactions in household flood insurance decisions.
- It uses over 50 million NFIP transaction-level observations from 2009 to 2019.
- The main outcome is county-month policies in-force, normalized to January 2010.
- Social connectedness is measured using Facebook county-pair friendship probabilities.
- The first quasi-experiment uses floods experienced by geographically distant friends.
- The second quasi-experiment uses FEMA flood-risk-map campaigns experienced by geographically distant friends.
- Distant friends' flood experiences raise local insurance demand by roughly 1% in baseline estimates and up to about 5% for significant floods.
- Distant friends' campaign experiences raise local insurance demand by about 1%.
- Effects are persistent, suggesting learning rather than temporary panic.
- Pre-trend tests and placebo tests support the causal interpretation.

- Migration, correlated local flood risk, generalized risk aversion, and other disasters do not explain the results.
- The most plausible mechanism is indirect social learning: friends' experiences trigger attention to flood risk and lead households to acquire local information.

8 Conclusion

8.1 Core conclusion

Hu shows that social interactions meaningfully affect household demand for flood insurance. Households increase purchases when geographically distant friends experience either floods or flood-risk-map campaigns. Because these shocks should not directly affect local flood risk or insurance supply, the results provide evidence of social-network transmission in insurance decisions.

8.2 Economic intuition

Flood risk is low frequency and easy to ignore. Many households have access to information about local flood risk and NFIP coverage but do not actively process it. When friends experience floods or campaigns, flood risk becomes salient. The household may then learn about its own exposure and buy insurance. Social networks therefore turn distant experiences into local financial decisions.

8.3 Policy implication

The results suggest that flood-risk communication can have network multipliers. Policymakers may increase take-up not only by subsidizing premiums but also by designing information campaigns that exploit social diffusion. However, the paper does not claim that higher NFIP take-up is always socially optimal because subsidy costs and government fiscal exposure must also be considered.

8.4 Final takeaway

The paper provides one of the clearest causal demonstrations that social finance matters for disaster insurance. People do not make insurance decisions in isolation: friends' experiences can shift attention, beliefs, and demand for protection against rare but severe risks.